Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations

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Abstract Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable — yet investigations unfold under closure pressure that structurally favors premature convergence on a single explanation before alternatives have been meaningfully compared. This paper reframes this failure as an HCI problem: not of generating explanations, but of maintaining comparison under conditions that prematurely collapse it. I introduce **Premature Accountability Convergence (PAC)** as a structural failure mode that locates the problem not in individual cognition but in the interaction between closure pressure, documentation requirements, and institutional incentives, and I propose *comparison preservation* as a design principle and *AI as comparison enforcer* as an inversion of the dominant model of AI in investigation support. A two-study empirical program with $N = 20$ GMP-experienced participants establishes the empirical basis and demonstrates the design response: Study 1 (timed reflection protocol at three timepoints) shows that 15 of 20 participants produced single-cause, person-centered explanations initially — with confidence averaging 1 out of 5 regardless of experience level, indicating institutional pressure rather than epistemic certainty — but 18 of 20 broadened their accounts within ten minutes; Study 2 (AI-supported RCA Reasoning Scaffolder with five structured stages) produced broader causal accounts in 18 of 20 participants, shifted 12 of 15 from person-centered to systemic explanations, and achieved usability ratings of 5.0 out of 5 and unanimous adoption intent, while all 20 participants simultaneously identified the audit trail implications of transparent investigation reasoning as a structural barrier to deployment.

Keywords: premature accountability convergence, root cause analysis, comparison preservation, AI scaffolding, GMP, closure pressure, regulated environments, human-AI interaction

Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable. An investigation into a deviation should, in principle, compare multiple plausible explanations before arriving at a documented conclusion. In practice, investigations unfold under closure pressure — deadlines, documentation requirements, and procedural expectations that define when an investigation

must be considered complete. Under these conditions, reasoning does not fail to generate alternatives; it fails to maintain them in a comparable state long enough for systematic evaluation. Convergence on a single explanation occurs not because other explanations have been eliminated, but because comparison between them has collapsed.

This paper reframes this dynamic as an interaction problem. The issue is not that investigators fail to generate alternatives, but that the workflow does not sustain those alternatives in a comparable state long enough for meaningful evaluation. Interface design determines which explanations remain visible and which are suppressed before comparison is complete — establishing the conditions of investigation reasoning as a domain in which HCI intervention is both possible and necessary.

I conceptualize this dynamic as **Premature Accountability Convergence (PAC)**: the early stabilization of causal accounts that occurs when the comparison between alternative explanations collapses before reasoning has fully developed. PAC is not treated here as a cognitive bias or individual reasoning failure. It is treated as a structural outcome — a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure. This framing shifts the design question from "how do we help investigators think better?" to "how do we design workflows that keep comparison open longer?"

I report a two-study empirical program in pharmaceutical quality systems. Study 1 establishes an empirical baseline using a timed reflection protocol with $N = 20$ GMP-experienced participants. Study 2 introduces an AI-supported RCA Reasoning Scaffolder that inverts the dominant model of AI in investigation support: rather than generating candidate explanations, the AI component detects convergence signals and requires their reopening — functioning as a comparison enforcer rather than an explanation source.

The contributions of this work are threefold. First, I introduce *comparison preservation* as a design principle for AI-supported reasoning systems in regulated environments. Second, I introduce *AI as comparison enforcer* as a design principle that inverts the dominant model of AI in investigation support. Third, I demonstrate empirically that comparison collapse is time-sensitive and addressable through design, and that structured scaffolding can delay convergence and broaden causal accounts.

Related Work

Causal Explanation as Organizational Outcome

The foundational premise of this study — that causal explanations in regulated investigations are shaped by the organizational conditions under which they are produced — draws on Vaughan's (1996) account of the Challenger launch decision. Vaughan shows that the engineers and

managers at NASA reached a catastrophic decision through a series of incremental, locally reasonable judgments structured by production pressure, communication patterns, and the normalization of prior deviations. Applied to GMP deviation investigations, this framework implies that the causal explanations produced by RCA reflect the conditions of deliberation, not only the evidence available. PAC names the specific mechanism through which this shaping occurs.

Recognition-Primed Decision Making and the Satisficing Problem

Klein's (1998) account of recognition-primed decision making (RPD) provides the cognitive background against which PAC becomes legible as a design problem. Klein demonstrates that expert decision makers under time pressure do not compare options — they recognize a situation, generate a single candidate course of action, and evaluate it before acting. In regulated investigation contexts, expertise accelerates the very dynamic that investigation quality requires resisting: the more experienced the investigator, the faster the initial convergence.

Sensemaking and the Structure of Investigative Reasoning

Klein et al. (2006) extend the RPD account through a sensemaking framework that treats diagnosis as the progressive construction of a plausible account from incomplete evidence. Sensemaking is inherently comparative — yet the conditions of regulated investigation systematically suppress this comparative structure. PAC is, in these terms, a premature termination of sensemaking.

Situated Action and the Conditions of Reasoning

Suchman's (1987) account of situated action establishes that reasoning is shaped by the immediate conditions of its execution. An investigation workflow that requires a documented root cause before alternatives have been meaningfully compared creates conditions that structurally favor premature convergence, independent of individual reasoning quality. The scaffold in Study 2 is a redesign of the situated conditions of reasoning.

Invisible Work and the Accountability Stakes of Comparison

Star and Strauss (1999) show that work performed in the gaps of formal systems tends to become invisible. In regulated investigation workflows, the comparative work of holding multiple causal explanations in parallel is precisely this kind of invisible work: not required by the documentation template, not rewarded by closure expectations, and not visible in the final investigation record.

Human-AI Interaction and Comparison-Preserving Design

The AI component of the Study 2 scaffold reflects Amershi et al. (2019)'s principle that AI systems should support human control and avoid pre-empting human judgment. In regulated investigation contexts, any system that generates causal conclusions risks introducing a compliance risk. The AI component is constrained to a specific function: detecting signals of premature convergence and triggering prompts that reopen the comparison space. Buçinca et al. (2021) demonstrate that cognitive forcing functions reduce over-reliance without sacrificing efficiency; the PAC detection mechanism extends this principle to a context where the risk is premature reliance on the investigator's own first account.

AI-Assisted Decision Making and Interface Effects

Wang and Yin (2021) demonstrate that explanation interfaces alter how decision makers engage with AI outputs, establishing investigation interfaces as active shapers of reasoning. Green and Chen (2019) show that algorithm-in-the-loop settings systematically fail to produce reliable decision making even when users are instructed to exercise independent judgment — grounding the design logic of comparison preservation: if interface structure shapes what explanations become salient, redesigning that structure is a tractable intervention against premature convergence.

Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation investigations in these environments are formal quality events requiring documented root cause analysis, CAPA planning, and qualified reviewer approval before closure — imposing closure pressure that shapes how reasoning develops before alternatives have been fully compared.

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team, with direct experience in GMP deviation investigation workflows including RCA, CAPA documentation, and quality review. Experience levels ranged from entry-level to senior practitioners with five or more years in GMP quality roles. The same-sample design eliminates between-participant variance in domain knowledge as a confound. Potential carry-over effects are addressed through a two-week minimum interval between studies, the scaffold's structural requirement to generate hypotheses not previously committed to, and the

fact that the primary outcome — a shift from person-centered to systemic attribution — is directionally opposite to what carry-over alone would predict.

Investigation Scenario

Both studies used the same standardized deviation scenario describing an out-of-specification result in an analytical testing context involving multiple instrument dependencies and procedural steps, designed to support multiple plausible causal interpretations across person-centered, procedural, equipment-related, and organizational categories.

Study 1: Timed Reflection Protocol

Study 1 established an empirical baseline for how reasoning develops under closure pressure without structured support. Participants produced causal explanations at three timepoints (0, $+5$, $+10$ minutes) using a standard RCA template, recording causal type (person-centered, procedural, equipment-related, organizational, or mixed) and confidence rating (1–5 scale). Causal type classification was performed directly by participants via structured response options, reflecting professionally grounded judgment consistent with GMP documentation practice.

Study 2: RCA Reasoning Scaffolder

Study 2 introduced an AI-supported RCA Reasoning Scaffolder implemented as a Streamlit web application, organizing reasoning into five sequential stages:

1. **Intake** (Figure 1): structured capture of deviation context without commitment to a causal conclusion. A live hypothesis panel tracks active, narrowed, and dropped counts alongside a closure signal indicator.

2. **Hypotheses** (Figure 2): generation of multiple causal candidates across person, procedure, equipment, environment, and organizational categories. Each hypothesis receives a PAC risk rating (High/Low) based on convergence-oriented language detection. Investigators are instructed not to narrow yet.

3. **PAC Analysis** (Figure 3): rule-based detection of convergence-oriented language with prompts to surface underexplored contributors. An AI component activates when individual-blame language is detected, surfacing alternative causal pathways and generating reopening questions.

4. **Reasoning** (Figure 4): explicit per-hypothesis evaluation of contributing factors and evidence, with plausibility rating and investigation note for each hypothesis before any single explanation is selected.

5. **Pre-Closure** (Figure 5): a required check confirming alternatives were considered before final attribution, displaying hypothesis state and posing three verification questions before closure is permitted.

The AI component was integrated into Stage 3 only, functioning exclusively as a conditional trigger for reopening the comparison space. It did not suggest causes, evaluate hypotheses, or generate conclusions.

Measures

Post-task assessment included five construct scales: usability (5 items), reasoning expansion (5 items), PAC mitigation (4 items), perceived usefulness (3 items), and blame pressure (1 item). All Likert items used a 5-point scale. Confidence change was assessed via pre/post 0–10 scale. Open-ended responses probed qualitative experience with the scaffold.

Findings

Study 1: The Time-Sensitivity of Comparison

At the initial timepoint, 15 of 20 participants produced single-cause, person-centered explanations. Confidence ratings averaged 1 out of 5 regardless of experience level, indicating that early convergence reflects institutional pressure rather than epistemic certainty. With five additional minutes, 8 of 20 revised substantially; by ten minutes, 18 of 20 had broadened their explanation or become more explicitly uncertain. Post-task discussion revealed that all 20 participants selected explanations oriented toward documentation acceptability rather than investigative completeness: "in a real investigation, you pick the explanation that closes the file. A systemic cause just opens more questions."

Study 2: Scaffold Effects

18 of 20 participants produced broader causal accounts with the scaffold. 15 of 20 reconsidered their initial attribution; of these, 12 shifted from person-centered to systemic or procedural explanations. Post-task assessment confirmed the scaffold's effects: usability averaged 5.0 out of 5; reasoning expansion averaged 4.8; perceived usefulness averaged 4.9; PAC mitigation averaged 4.7; blame pressure averaged 1.0 out of 5. Confidence ratings were maintained or slightly decreased while causal breadth increased substantially — a dissociation indicating that comparison-preserving scaffolding does not produce epistemic destabilization. In high-accountability regulated contexts, expanded reasoning without confidence collapse is precisely the outcome that makes adoption feasible.

The usability ceiling reflects a domain effect rather than a measurement artifact: GMP investigation workflows have no existing analogue for structured comparison scaffolding, and participants evaluated the tool against the absence of any structured support. All 20 participants expressed adoption intent while simultaneously identifying the audit trail as a structural barrier to deployment: "the problem isn't the tool. The problem is that the tool would show that we looked at the systemic cause and decided not to pursue it. That's a conversation no one wants to have."

Discussion

Comparison Preservation as Interaction Design

Comparison preservation is a structural intervention, not a cognitive aid. It maintains the availability of alternative explanations as an interaction condition. Prior HCI work establishes that interface structure actively shapes human judgment (Wang and Yin, 2021; Green and Chen, 2019) and that cognitive forcing functions reduce over-reliance (Buçinca et al., 2021). Comparison preservation extends this work into a more specific context: the risk is not over-reliance on AI outputs but over-reliance on the investigator's own first account.

AI as Comparison Enforcer

The dominant model for AI in investigation support treats AI as a producer of candidate explanations that narrows the comparison space. The scaffold inverts this logic: the AI component's function is to detect when comparison has collapsed and to require its reopening. This instantiates Amershi et al.'s (2019) judgment-preservation principle in a more demanding form: in a high-accountability regulated context, preserving human judgment means designing AI to resist institutional pressures that produce premature convergence.

PAC as a Structural Problem

The finding that all 20 participants selected explanations oriented toward documentation acceptability rather than investigative completeness is evidence of rational adaptation, not poor reasoning. Confidence ratings averaging 1 out of 5 at the initial timepoint establish that convergence is produced by institutional structure, not by belief. This extends Vaughan's (1996) account: the explanation that gets documented is shaped by the accountability environment in which the investigator operates.

The Organizational Adoption Problem

All 20 participants expressed adoption intent; all 20 simultaneously identified the audit trail as a structural barrier. This combination reflects the gap between individual preference for investigative completeness and organizational incentive structures that reward documentation efficiency. The scaffold surfaces a condition that currently remains invisible: the choice not to pursue a systemic explanation is already being made, just without a record. Whether organizations can tolerate that legibility is an organizational and regulatory question, not a design one — but it is a question that Regulatory HCI is positioned to name.

Limitations and Future Work

The use of $N = 20$ participants from a single GMP quality team reflects a structural constraint of the domain rather than an arbitrary methodological choice — competencies required for GMP investigation work are not transferable from adjacent populations, and $N = 20$ represents the boundary of what domain-valid sampling permits. Future work should examine PAC in live investigation settings and investigate the organizational conditions under which comparison-preserving scaffolding can be deployed.

Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a structural failure mode in regulated investigation workflows and reframed it as an interaction problem concerning the conditions under which comparison is sustained or collapsed. Study 1 established that comparison is time-sensitive and that early convergence reflects institutional pressure rather than epistemic certainty. Study 2 demonstrated that an AI-supported RCA Reasoning Scaffolder — with an AI component functioning as a convergence detector rather than an explanation source — produced broader causal accounts, shifted attribution from person-centered to systemic explanations, and achieved unanimous adoption intent. Three contributions follow: comparison preservation as a design principle, AI as comparison enforcer as a design inversion, and the organizational adoption problem as a first-class research concern for Regulatory HCI.

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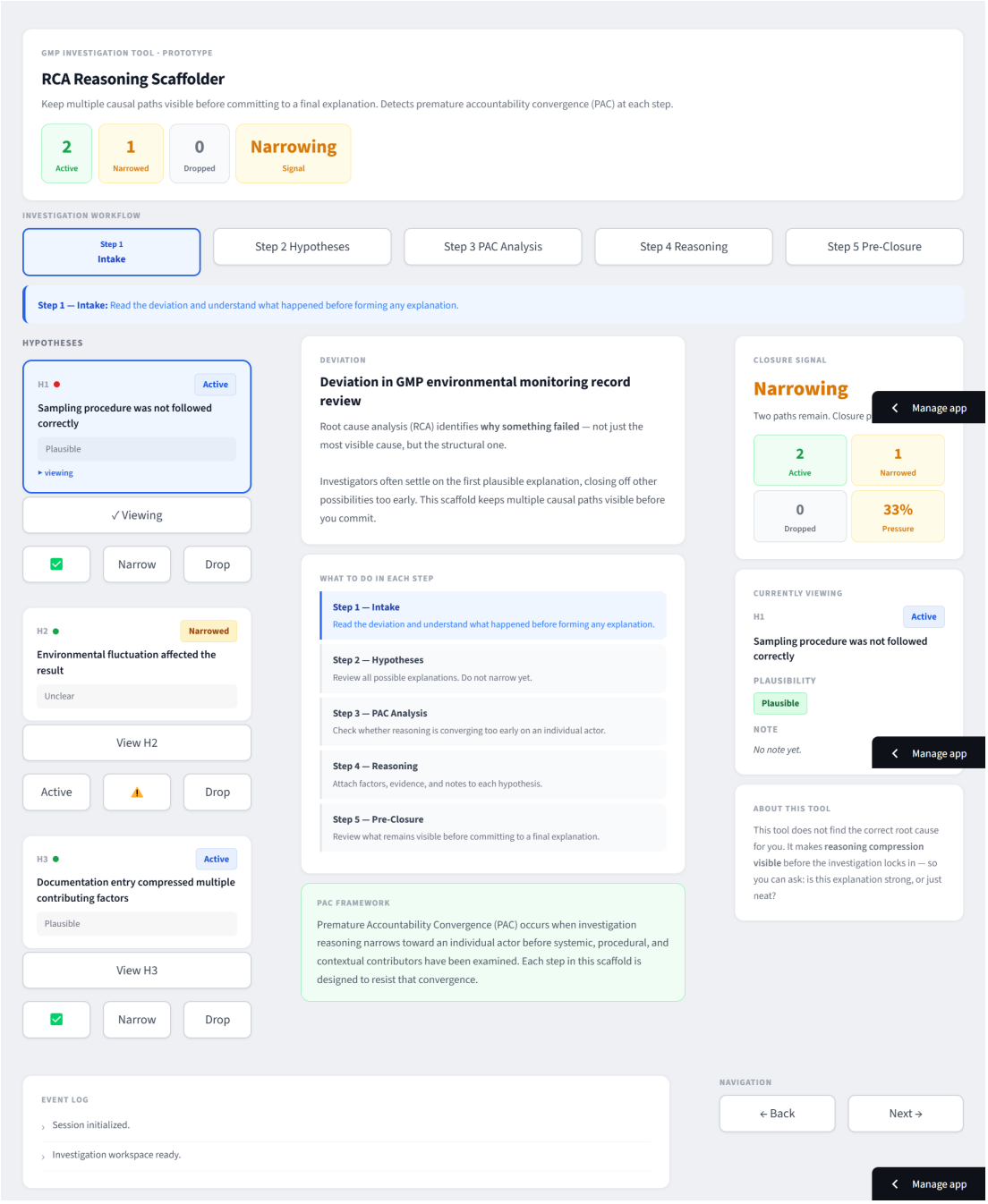


Figure 1: RCA Reasoning Scaffolder: Step 1 (Intake). The hypothesis panel shows three active hypotheses. The closure signal panel shows 2 active, 1 narrowed, 0 dropped, 33% pressure — indicating narrowing has begun before hypothesis expansion is complete.

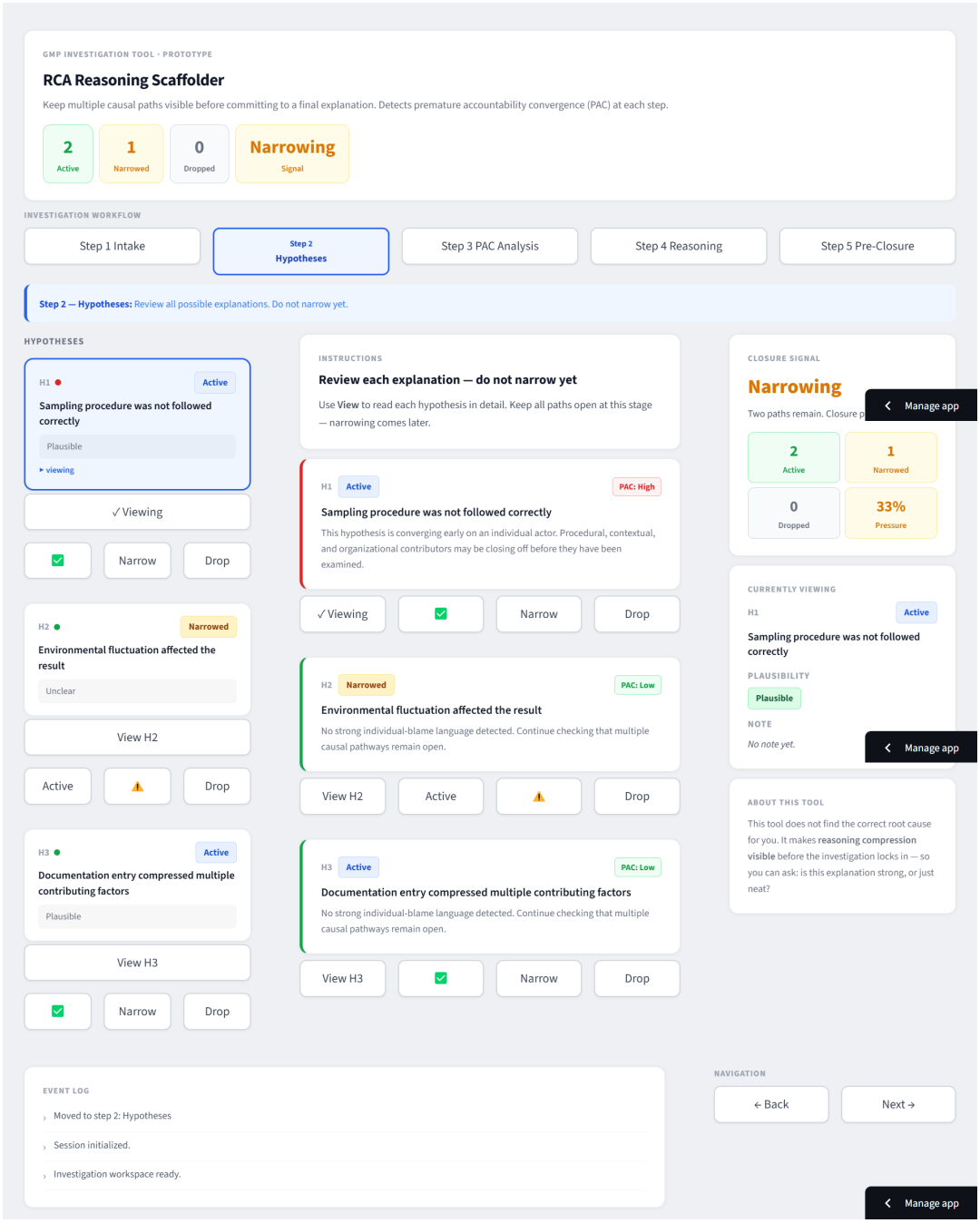


Figure 2: RCA Reasoning Scaffolder: Step 2 (Hypotheses). H1 is flagged PAC: High, indicating convergence-oriented language targeting an individual actor. The instruction "Review each explanation — do not narrow yet" enforces comparison preservation at the stage level.

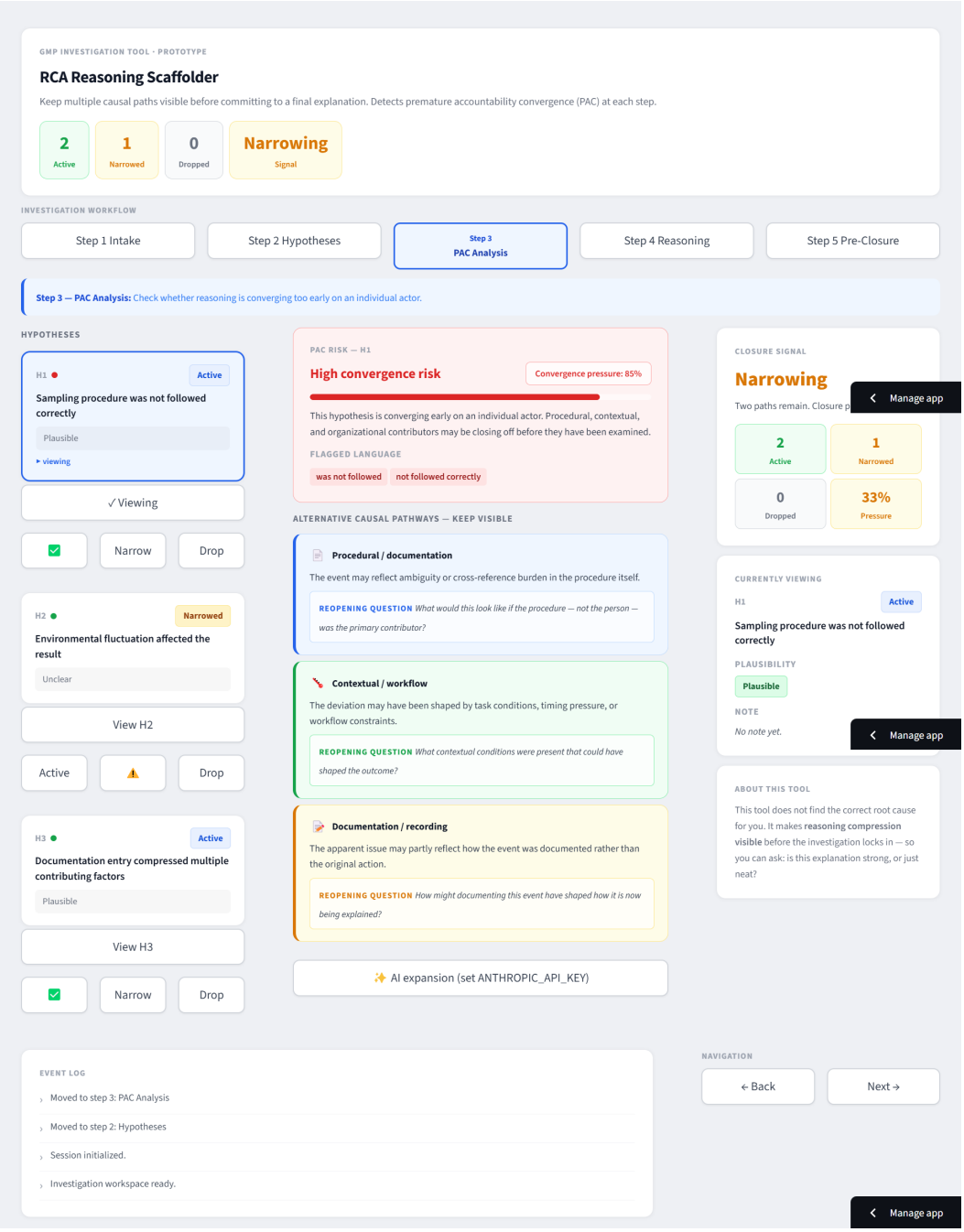


Figure 3: RCA Reasoning Scaffolder: Step 3 (PAC Analysis) with AI layer active. H1 is flagged as high convergence risk (85%) with individual-blame language tokens highlighted. The AI component surfaces three alternative causal pathways with reopening questions that must be considered before closure.

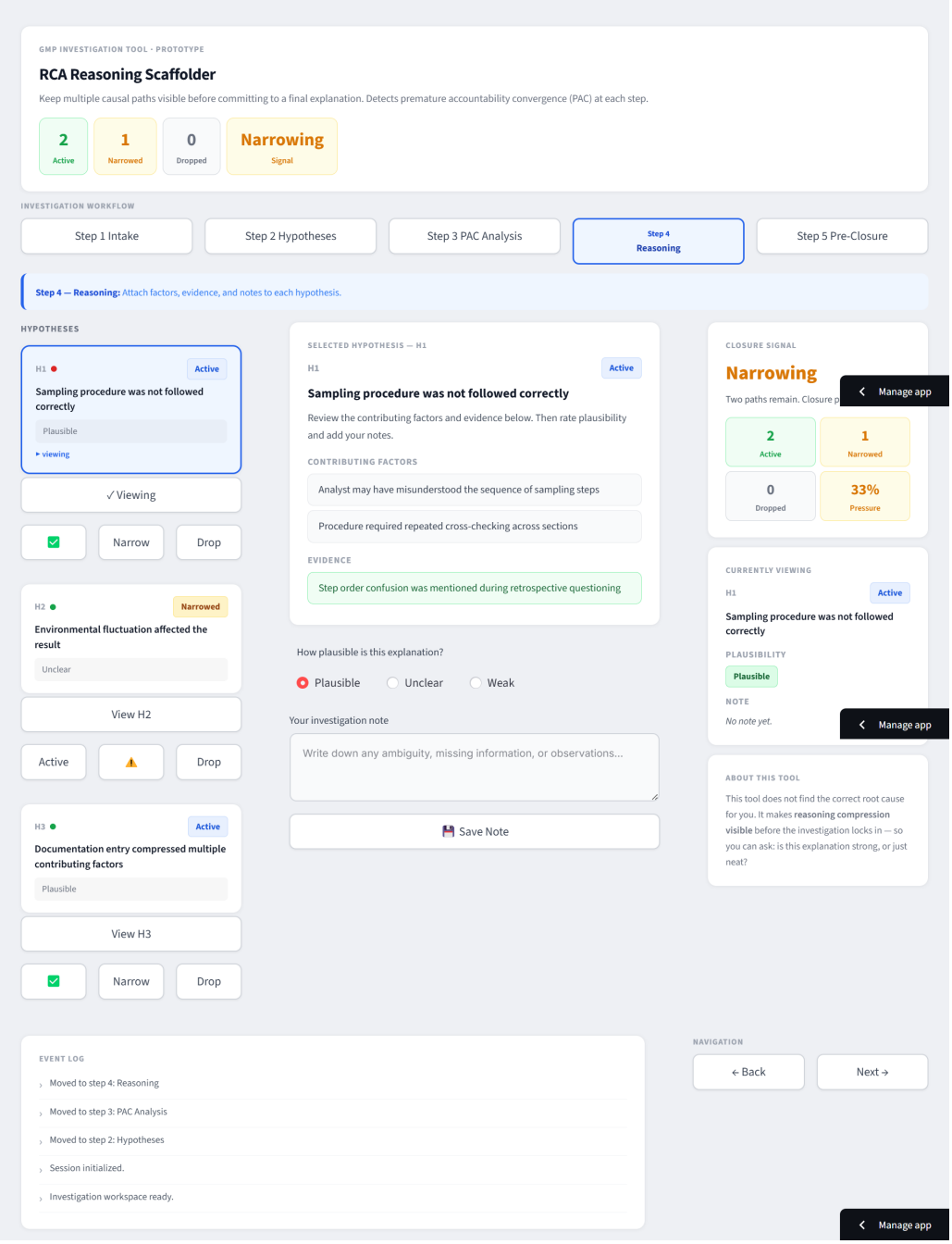


Figure 4: RCA Reasoning Scaffolder: Step 4 (Reasoning). Contributing factors and evidence for H1 are displayed for explicit evaluation. The investigator rates plausibility and adds investigation notes before moving to the next hypothesis.

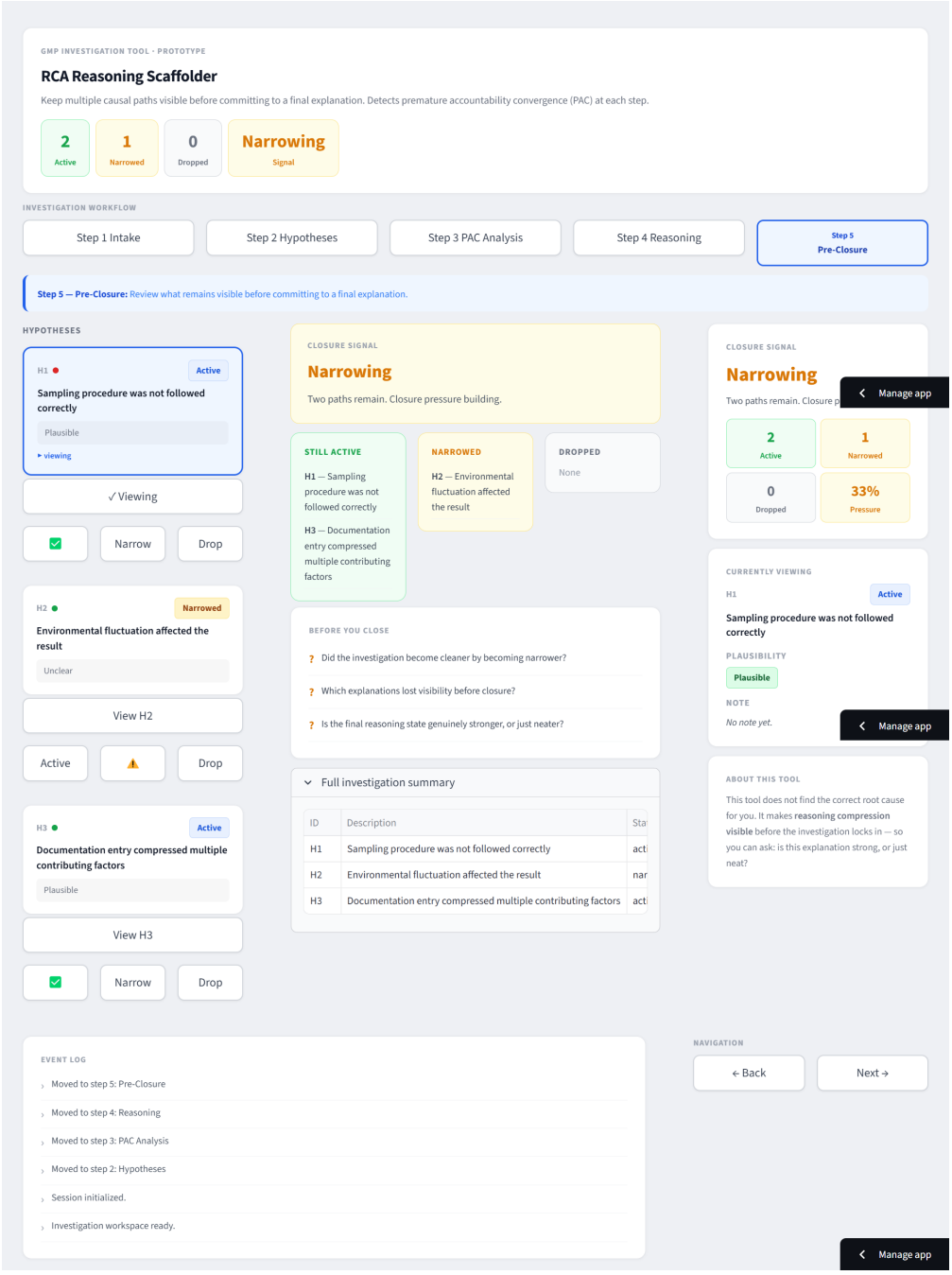


Figure 5: RCA Reasoning Scaffolder: Step 5 (Pre-Closure). Three verification questions are posed before closure is permitted: “Did the investigation become cleaner by becoming narrower?”, “Which explanations lost visibility before closure?”, and “Is the final reasoning state genuinely stronger, or just neater?”